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Embodied Intelligence Dexterous Hands (EIDH) Brief Information Report

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1. Introduction

Dexterous Hands are high-degree-of-freedom (DOF) robotic end-effectors that mimic the structure and function of the human hand. Through intricate biomimetic structures and actuation methods, they simulate the flexible movement and fine control of human hands, possessing the potential to perform human-like operations.

1.1. Basic Technical Features and Classification

High Degree of Freedom (DOF): DOF is a core metric measuring the flexibility of dexterous hands. The human hand has between 12-17 DOF. Current top-tier dexterous hands, like the Tesla Optimus hand, has 22 DOF. Yushu Technology's Dex5 had integrates 20 DOF (16 active + 4 passive).

Actuation and Transmission Methods: There are two main categories: **Fully actuated** (high precision but high cost) and **Underactuated** (simple structure and lower cost). Transmission methods are mainly **Tendo-drive** (e.g., Tesla, high flexibility) and **Linkage-driven** (mature and commonly used), others are like gear transmission. Each scheme has its advantages and disadvantages; for instance, miniature lead screw transmission offers high efficiency and high integration but at a higher cost, tendon-driven transmission offers high space utilization¹ and flexibility but lower lifespan and reliability, and linkage-driven transmission provides high load capacity and precision but is unsuitable for high-speed motion.

Sensing Capability: Tactile perception is key for dexterous hands to adapt to complex environments. Without sensors, a robotic hand is "blind" and "numb". It enables autonomy and reflexes. Sensors provide the 'local intelligence' needed for the hand to have 'reflexes'. This is often handled by lower-level control loops that don't require the main 'brain' (the central processor) to micromanage every millisecond. This is called closed-loop control.

Materials and Manufacturing: To balance performance, weight, and cost, dexterous hands utilize materials like alloys, plastics, and 3D printing. The USTC's 19-DOF biomimetic dexterous hand weighs only 0.37 kg [1], achieving a balance of wearability, high degrees of freedom, and precise control.

Control and High-Level AI: One of the key aspects of dexterous hand is its control system, as for the basic function of that, is to mimic human's hand to grab and transfer object or achieving tasks like toughing. For reaching that goals, training is required for flexibility, so as its control system as that is one of the key elements for doing that. But reaching higher levels of dexterity is not easy and one of the reasons is due to

¹ **Space Utilization** refers to the efficient packaging of all necessary components (actuators, sensors, transmission systems, wiring, and structural elements) within the physical volume of the hand and forearm, maximizing functionality and performance while minimizing the overall size and weight.

computational sensorimotor learning limitations [2]. For higher Degree of Freedom (DOF) the challenge of complexity is also higher. There are two major ways for training: one is teleoperation, the other one is through Reinforcement Learning (RL) or Imitation Learning (IL).

1.2. Report Objectives and Scope

This report is going to shortly analyse the development of the EIDH based on some key elements and deep go through three of them. Comparison would be made and listing the advantages and disadvantages of them so that to get a better view of the current development of EIDH. Marketing and industry will also be looked for better analyse. Based on these analyse, the challenges and difficulties will become as clear as possible and therefore, understand the target and direction for us to further analyse and develop.

2. The Review on dexterous hands

The development of human-like robot end-effector is to mimic human hands as close as possible, like the flexibility, adaptability, and complexity. Grasping serves as the fundamental interaction between the robot and its environment, enabling object manipulation. This action involves contact between both parts and usually this operation also requiring no damage for both as well [3]. Therefore, to do achieve that, delicate control of the gripper is necessary because serious movements are based on the control system to “instruct” each part of the gripper; sensor is also important, just like human hands, which are capable to feel the environment like temperature, texture of the object, or the weight. Study shows that tactile sensors can improve the accuracy for dexterous hands [4]; material used for gripper is another element need to consider as for grip object for example, object should not be damaged or slid out of it, and material for gripper should be durable for long-term operation and yet not affect the flexibility. Another reflection of flexibility is though the Degree of Freedom (DOF) which is the number of independent movements that can be achieved. For human hands, it has 27 of DOF [5]. Robot gripper started with two fingers but commonly with three and Okada dexterous hand designed in 1979, Stanford/JPL hand and the Utah/MIT hand in early 1980s were the representatives of dexterous hands. They have three fingers with DOF varies between 9-25. In 1990s NASA was designed robotic dexterous hands for space operation: Robonaut hand [6]. These dexterous hands have 5 fingers with 14 DOF. Nowadays, companies like Tesla or Yushu (Unitree) are providing innovative design for dexterous hands with higher DOF, integrated censoring capabilities, tendo-driven architecture [7] and reverse drive [8].

All of these show a clear development history of robotic dexterous hands through DOF, Sensors, Transmission and Drive. In the following contents, deep investigation would be carried out based on several aspect, include but not limit to mechanical design, transmission and motors.

3. Core Technology Analysis

3.1. Mechanical Design

Mechanical design of dexterous hand is a Trade-off between Adaptability, Coupling and Rigidity. It is the mechanical structure that provides the connection between the environment and the hand.

In terms of adaptability, underactuated and fully actuated designs represent two different paradigms. Underactuated designs employ simple mechanical structures, enabling passive adaptation to the environment. Through a simple control system, they can conform to the geometry of objects, achieve stable grasping (locking), and proceed with the next operation. This inherent simplicity means that the structural design of underactuated end-effectors is more challenging than that of fully actuated designs, directly leading to reduced manufacturing and maintenance costs[9]. Therefore, this trade-off between simplicity and cost-effectiveness results in relatively lower operational precision. This is because fewer motors are used, and each joint cannot be independently controlled to achieve precise angular motion.

The primary advantage of a fully actuated hand is its high accuracy and precision, achieved by placing a dedicated motor at each major joint. However, this design pattern comes with higher manufacturing and maintenance costs. In terms of adaptability, fully actuated hands contrast with underactuated hands. They lack the inherent, passive adaptability to unknown object shapes. Therefore, to achieve robust grasping and manipulation of different objects, complex control strategies are required, typically developed through extensive simulation and training. For example, performing precise in-hand manipulation, such as picking up a pen with two fingers or gently rotating an object, is a core advantage of fully actuated systems. These tasks often exceed the capabilities of underactuated hands, as the mechanical simplicity of underactuated hands sacrifices fine control in favour of robust, passive grasping.

Aspect	Underactuated	Fully actuated
Complexity and cost	Lower (fewer motors, simple control)	Higher (motors for major joints)
Control	Simple (command for gasp/open only)	Complex (required precise coordination of all joints)
Adaptability	High passive adaptability	Low passive adaptability
Robustness	High (simpler mechanism, less to break)	Can be lower (more complex components)
Precision	Lower (cannot control every joint angle directly)	High (direct control over every joint)
Primary Goal	Robust, stable grasping of a wide variety of shapes	Precise, in-hand manipulation

Table 1, comparison between underactuated and fully actuated

Coupling is a fundamental architectural decision in robotic mechanism design, determining the relationship between independent degrees of freedom. In a fully decoupled system, each joint or actuator operates independently, enabling direct and precise control over each independent motion. This decoupling is crucial for precision manipulation tasks requiring independent finger gaits or specific force applications with a single finger. Conversely, coupled designs intentionally link the motion of multiple joints through mechanical elements such as links, tendons, or differential mechanisms. This approach is central to underactuated technologies, which sacrifice independent control to achieve significant advantages like weight reduction, cost reduction, and inherent passive adaptability. For example, a coupled finger can automatically conform its entire motion chain to the shape of an object upon contact, without any sensor input or complex control algorithms. Thus, the engineering trade-off lies in balancing the control purity provided by decoupling with the simplicity, robustness, and adaptive intelligence brought by strategic mechanical coupling.

Stiffness balance represents a fundamental design trade-off in robotic systems, determining the compromise between structural rigidity and flexible adaptability. High-stiffness designs are characterized by robust materials and fixed joints, offering excellent precision, stability, and power transmission, which are crucial for tasks such as high-speed assembly or heavy object handling. However, this stiffness comes at the cost of environmental fragility; such systems struggle to absorb accidental impacts or interact safely with fragile objects and humans. Conversely, systems designed for compliance intentionally introduce flexibility through soft materials, passive joints, or series elastic actuators. This design philosophy grants them exceptional robustness and passive safety, enabling them to adapt to irregular object grasping and maintain stable contact with uncertain environments. The core challenge of "stiffness balance" is therefore strategically distributing stiffness and compliance within the mechanism to achieve performance for specific tasks while not compromising overall reliability and safety.

Mechanical design with dexterous hands is essentially a trade-off between the three core elements of adaptability, coupling, and rigidity. Underactuated design achieves passive adaptability and robust grasping through mechanical coupling, at the cost of sacrificing precision; while fully actuated design realizes high precision and dexterous operation through joint decoupling but introduces higher complexity and cost. Finally, rigidity balance determines the allocation between structural stiffness and compliance, representing a critical choice between safety in environmental interaction and precision in task execution. In summary, there is no single optimal design solution; the best choice entirely depends on the robot's intended tasks and application scenarios.

3.2. Transmission

The design of the transmission system is a key factor determining the performance and characteristics of dexterous robotic arms. It serves as the critical connection between the actuator (power source) and the joints executing motion. The choice of transmission mechanism directly impacts the robot arm's weight, compactness, output force, and reversibility [10]. Rigid transmissions, such as high gear ratios or linkages, provide high fidelity in position control and exhibit excellent force transmission performance, making them ideal for precision, high-stiffness tasks. However, this often comes at the cost of increasing the weight and complexity of the robot arm's structure. In contrast, cable-driven systems offer notable advantages by using cables to transmit force from remote locations, significantly reducing the robot arm's weight and inertia, allowing designs to approach human-like forms and agility. Nevertheless, this method introduces challenges such as friction, cable stretching, and the need for continuous tensioning, which may affect control accuracy and long-term reliability. Ultimately, transmission design embodies a fundamental trade-off: positioning the actuator to achieve optimal power and control versus minimizing moving mass to enhance speed and safety. Each configuration shapes the final capabilities and applicability of the hand.

Aspect	Tendon-Driven	Linkage/Gear-Driven
Core Principle	Transmits force and motion remotely via cables (tendons), similar to biological tendons	Transmits force and motion directly through rigid components (gears, linkages)
Actuator Placement	Actuators can be remotely located, drastically reduction the weight of the end-effector	Actuators are typically mounted at or near the joint, increasing the weight and inertia of the moving parts
Precision & Stiffness	Lower, easy to elastic stretch, friction, hysteresis, and lower overall stiffness	High, offers high rigidity, high transmission efficiency, low backlash, and excellent positional accuracy
Back drivability	Generally poor	Generally good
Force/Tactile Sensing	Joint torque can be estimated indirectly by measuring tendon tension, through friction can corrupt the signal	Typically requires force/torque sensors to be mounted directly on the joint, adding complexity
Inherent Compliance & Safety	Inherently compliant, elasticity o tendons can absorb shocks, making interactions safer	Inherently rigid, achieving compliance requires complex active force control algorithms

Complexity & Maintenance	Complex, require frequent maintenance	Structurally complex but highly reliable, requires less frequent maintenance
Typical Applications	Dexterous hands, lightweight robot arms, bio-inspired robots	Industrial robot arms, high-precision CNC machines, articulated robots

Table 2, comparison between tendon-driven and linkage/gear-driven

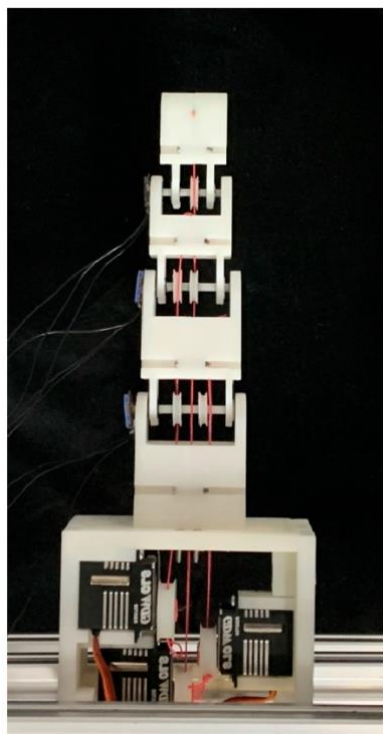


Figure 1, Demonstration of tendon-driven finger[11]

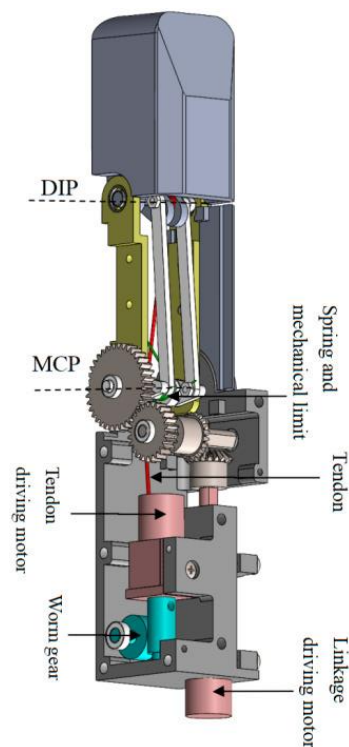


Figure 2, Demonstration of Linkage/gear-Driven finger[12]

3.3. Motors

The configuration of the drive motor is the main determinant of the dexterous robotic arm's mechanical design, directly controlling the trade-off between adaptability, coupling, and rigidity. The number, location, and control methods of these motors define the basic capabilities of the arm. Here are some most common configurations with advantages and disadvantages for them:

1. Rotary Motor + High-Ratio Gear Reducer (e.g., Planetary Gearbox)

This is one of the most common combinations in robots. The rotating motor (usually a brushless DC motor or servo motor) provides high-speed rotation, which is then converted to high torque at a lower speed through a planetary gear box.

Advantages: Achieved very superior torque-to-weight ratio, compact size, and high efficiency. It strikes a good balance between precision and power, making it suitable for joints requiring significant strength.

Disadvantages: Usually, due to friction and gears, the low reverse drive capability causes the joint to feel "stiff" and complicates passive compliance. It may also introduce audible noise.

2. Rotary Motor + Worm Gear

This combination uses a worm screw to drive a worm wheel, offering a unique set of properties.

Advantages: Provides a high reduction ratio in a single stage to achieve a compact design. Its most notable feature is the built-in self-locking function, which is crucial for holding position without power (e.g., in grippers). This enhances the safety of vertical applications.

Disadvantages: Inherently inefficient, it exhibits very poor performance under reverse driving. The sliding friction of this mechanism also generates more heat [13], which could be a limiting factor.

3. Direct Drive

This configuration eliminates the transmission entirely by connecting the load directly to the motor rotor.

Advantages: Provides perfect back drive capability, zero backlash, and extremely high bandwidth control. This brings excellent force control fidelity, transparency, and the ability to safely absorb impacts.

Disadvantages: A large and heavy motor needs to be equipped for the given torque output. This is usually the most expensive solution and can lead to cumbersome joint designs, making it difficult to implement in compact spaces like finger joints.

4. Linear Motor + Tendon Drive

This is a hybrid approach where a linear actuator (or a rotary motor with a capstan) pulls on cables (tendons) to actuate remote joints.

Advantages: It allows the motor mass to be decoupled from the moving limb. This significantly reduces inertia, making the end effector (such as fingers) very compact and lightweight, thereby enabling high-speed and agile movement.

Disadvantages: Introducing nonlinear factors, such as cable stretching, friction, and hysteresis, which complicate control and reduce position accuracy. It also requires a tension mechanism and is prone to wear.

There are no “perfect” solutions for motor selection or combination, it is a result of combination of series of standards. Under different circumstances or environments, the selection of motors can be varied for targets of precision, torque, force, lightweight etc. Therefore, it is crucial to define the requirements and using comparison methods or through experiments to select the “perfect” motors for the dexterous hands.

4. Marketing

4.1. Marketing scales and foresee future

The global dexterous hand market is experiencing rapid growth, mainly due to the accelerated development of the humanoid robot industry. Dexterous hands were once a niche field within academia and research institutions but are now becoming a key component of the next generation of automation.

Current market analysis, as shown by data from institutions such as MarketResearchFuture and QYResearch, indicates a significant upward trend. In 2024, the global multi-finger dexterous robot market is valued at approximately \$2.02 billion. The expected compound annual growth rate (CAGR) is projected to be above 26% to 39.2%, with the market size reaching around 15.26 billion by 2030 [14]. The huge differences in the forecast highlight that although the market is in its early stages, it has explosive potential, which is highly dependent on the timeline of humanoid robot applications.

The Chinese market also reflects this growth. The Chinese humanoid robot industry is expected to expand from approximately 5.3 billion yuan in 2025 to 387 billion yuan in 2028 [15]. As a core high-value component, the value of a dexterous hand can constitute over 30% of the total cost of humanoid robots, highlighting its economic importance in the supply chain.

4.2. Competitors Analysis

The fiercely competitive dexterous hand market includes mature international players, agile local startups, and vertically integrated component manufacturers. They can be divided into two major categories:

1. Professional dexterous hand manufacturer:

- **International:** Companies like **Shadow Robot Company (UK)** and **SCHUNK (Germany)** represent high-end, mature players. They offer high degrees of freedom, research-grade hands, known for their precision and reliability, but usually at a higher price.
- **Domestic (China):** This section is highly dynamic and innovative.
 - **Inspire Robots(因时机器人):** A key participant that has developed its own core micro-linear servo actuators, providing complete robotic arms and critical components to other robot manufacturers.
 - **PaXini (帕西尼):** Characterized by its focus on high-density, multimodal haptic sensing technology, it integrates hundreds of haptic sensors into its hand-shaped design to achieve exceptional environmental interaction.

2. A humanoid robot company with independent hand research and development capabilities:

- Tesla (USA): Its Optimus Gen 3 hand, with 22 degrees of freedom and integrated tactile sensors, sets a benchmark for performance and cost reduction goals, driving industry expectations.
- Unitree Robotics (China): The launch of its Dex5 hand, with 20 degrees of freedom and reverse driving capability, showcases how a humanoid robot company vertically integrates to control key technologies.

5. Challenges and Difficulties with potential Develop and Analysis aspects

Although dexterous hand technology is developing rapidly and has broad market prospects, it still faces a series of core challenges in its technological realization and commercialization.

Technical bottlenecks and industrialization difficulties:

1. The efficiency and reliability of the transmission system: issues such as sliding friction and heat generation in worm gears, as well as cable stretch, friction, and wear in tendon drives, directly impact the precision, lifespan, and reliability of the dexterous hand. Improving the durability of the transmission system while maintaining performance is a major engineering challenge.
2. High-density sensing and integration: integrating a large number of tactile and force sensors into the narrow finger space and solving the resulting signal wiring, anti-interference and real-time data processing problems are the key to achieving true “tactile intelligence”.
3. Control complexity: the kinematic and dynamic models of a fully actuated, high-degree-of-freedom dexterous hand are extremely complex. Achieving human-level precision manipulation (such as rotating an object in the hand) requires solving real-time planning and control problems in high dimensional states, which places extremely high demands on both algorithms and computing power.
4. Cost-performance trade-off: the various drive and transmission solutions compared in the report are essentially a trade-off between performance, cost, and complexity. Producing a stable and reliable dexterous hand that meets the needs of specific application scenarios (such as industrial sorting or home services) within an acceptable cost range is a prerequisite for large-scale commercialization.

Potential research directions and development areas:

- i. New transmission and drive solutions: explore drive methods based on smart materials (such as shape memory alloys and hydraulically amplified self-healing electrostatic actuators[16]), or develop new low-friction, high-rigidity composite transmission mechanisms to fundamentally improve transmission efficiency, weight and reliability.
- ii. Bionic design and smart materials: deeper research into the anatomical structure of the human hand will lead to the development of dexterous hands with variable-stiffness joints and biomimetic tissues. Leveraging the properties of smart materials, dynamic adjustment of passive adaptability and active stiffness can be achieved.

- iii. Soft-robot fusion: combining the inherent flexibility and safety of soft robots with the precision and strength of rigid structures to develop a flexible and rigid-coupled dexterous hand may be an effective way to break through the current difficulty in balancing adaptability and rigidity.
- iv. Modular and standardized design: promoting the modularization and interface standardization of key components of dexterous hands (such as micro-actuators and tactile sensors modules) will help reduce research and design and manufacturing costs, accelerate the establishment of an application ecosystem.

6. Conclusion

Dexterous hands, as the core terminal for the interaction between robot technology and the environment, embody the collaborative progress of multiple fields such as mechanical design, transmission technology, drive systems, sensing and control. This report indicates through analysis that there is no one "perfect" universal design scheme, whether it be the driving method (full drive and underdrive), transmission scheme (tendon drive and linkage/gear drive), or specific motor configuration; their essence is a series of engineering trade-offs for specific application scenarios.

Future development will largely depend on breakthroughs in core technologies (such as solving challenges of transmission reliability and control complexity) and successfully finding the optimal balance between cost and performance. With the continuous integration of artificial intelligence and hardware technology, dexterous hands are expected to gradually transition from laboratory and research equipment to becoming a key component that empowers the next generation of intelligent robots, ultimately achieving universal operational capabilities comparable to human hands.

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